Relative error for an angular SxPID3 atom

A uniform eight-term bound and a correction to the earlier report

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Hand-derived population result; formal verification UNRUN. For the fixed law below, eight even-moment terms give

$$0 \le \frac{A - A_8}{A} \le \frac{1}{1253376} < 10^{-6} \quad \text{when} \quad 0 < |a| + |b| \le \tfrac{1}{2}.$$

Here A is the averaged top atom of the stated continuous functional. The proper terms in A_8 are exact real quantities. This is not a floating-point, sampling, estimator-calibration or application guarantee. At the origin, $A = A_8 = 0$; the relative quotient is undefined.

1 Correction and population model

The earlier *Conditional angular references for continuous SxPID3* report says "zero ratios" on page 4. For its phase-sum proper nodes, the correct statement is

$$G_\alpha/(p_T F_\alpha) = 1, \qquad i_\alpha = \ln 1 = 0, \qquad R_\alpha = 0.$$

The ratios are one; their logarithms are zero. This supplement preserves that report and its historical execution hold. It adds the following two-harmonic bound, which the earlier source did not supply.

Put $L = 2\pi$. The variables S_1, S_2, S_3, T are scalar angles in $[0, L)$, in radians, with fixed product Lebesgue measure. Fix real a, b, let $r = |a| + |b| < 1$, and define

$$v = (t - s_1) \bmod L, \qquad u = (t - s_1 - s_2 - s_3) \bmod L,$$
$$p(s_1, s_2, s_3, t) = L^{-4}(1 + a\cos v + b\cos u).$$

Its positive density lies between $(1 - r)L^{-4}$ and $(1 + r)L^{-4}$. Integrating t removes both harmonics, giving $p_S = L^{-3}$; integrating s_1 gives $p_T = L^{-1}$. Thus the law is normalized, the sources are independent uniform angles and the target is uniform. The torus density is smooth, and coordinate seams have probability zero. The bounded positive likelihood ratios make the information integrals finite. All logarithms are natural; information is in nats.

2 The functional and its proper averages

The defining method is the pure-continuous branch-sum construction of Ehrlich et al., arXiv v3, Sections II.2 and III.2, Equation (16), and Appendix B Definition 3. The model and the error bound in this report are project-derived. They are not theorems attributed to that paper. Multiply its bit values by $\ln 2$ to obtain nats. Categorical MGW and the general RCP/RN construction are different objects.

For an antichain α of incomparable nonempty source subsets,

$$F_\alpha = \sum_{B\in\alpha} p_{S_B}, \quad G_\alpha = \sum_{B\in\alpha} p_{T,S_B}, \quad i_\alpha = \ln \frac{G_\alpha}{p_T F_\alpha}, \quad R_\alpha = \mathbb{E}_p i_\alpha.$$

These branch sums are sums of densities, not finite-union probabilities. The same population law supplies every marginal and every average. The fixed radian measure also applies when branch sizes differ; normalizing each marginal to a unit-Haar density changes the functional.

For any nonempty proper source subset B, an omitted source removes the b harmonic. The remaining marginals are

$$p_{S_B} = L^{-|B|}, \qquad p_{T,S_B} = L^{-|B|-1} \begin{cases} 1 + a\cos v, & 1 \in B, \\ 1, & 1 \notin B. \end{cases}$$

Every proper node therefore has local ratio $1 + aw_\alpha \cos v$, where

$$w_\alpha = \frac{\sum_{B\in\alpha,\, 1\in B} L^{-|B|}}{\sum_{B\in\alpha} L^{-|B|}}, \qquad 0 \le w_\alpha \le 1.$$

Under the actual law, v has density $(1 + a\cos v)/L$. Define

$$R(w) = \frac{1}{L}\int_0^L (1 + a\cos v)\ln(1 + aw\cos v)\, dv,$$
$$P = R(1/2), \qquad D = R(2/3), \qquad Q = R(1).$$

The averaging weight remains a, not aw. For $|c| < 1$,

$$H(a,c) = \ln\frac{1+\sqrt{1-c^2}}{2} + \frac{ac}{1+\sqrt{1-c^2}}, \qquad R(w) = H(a, aw).$$

To obtain this expression, set $z = c/(1+\sqrt{1-c^2})$ and $K = (1+\sqrt{1-c^2})/2$. Then $1 + c\cos v = K|1 + ze^{iv}|^2$, with $|z| < 1$. Its absolutely uniformly convergent log series has uniform mean $\ln K$ and cosine-weighted mean z. The lower-bound proof below can also use the integral directly.

3 The top cut and the positive series

Write $\beta \preceq \alpha$ if every branch of α contains a branch of β. The signed averaged atoms satisfy $R_\alpha = \sum_{\beta \preceq \alpha} \Pi_\beta$. There are 18 nodes. The unique top is $\{123\}$.

Every proper antichain has a branch that extends to a pair, so the lower sets of $\{12\}, \{13\}, \{23\}$ together cover exactly the seventeen proper nodes. Intersections of these lower sets are the lower sets of the antichains containing the corresponding pairs. Inclusion-exclusion of their signed atom sums gives

$$\begin{aligned}\sum_{\text{proper } \beta} \Pi_\beta = R_{\{12\}} + R_{\{13\}} + R_{\{23\}} \\ - R_{\{12\}\{13\}} - R_{\{12\}\{23\}} - R_{\{13\}\{23\}} \\ + R_{\{12\}\{13\}\{23\}} = Q + D - 2P.\end{aligned}$$

The seven weights are $1, 1, 0, 1, 1/2, 1/2, 2/3$: all branches in these cuts have size two. No positivity of individual atoms is assumed.

Use $h = L^{-4}$ only as a uniform probability reference for calculation. The torus change to (s_1, s_2, v, u) has integer inverse $t = v + s_1$, $s_3 = v - u - s_2$ and determinant magnitude one. Hence v, u are independent uniform angles under h. With $Y = a\cos v + b\cos u$, full MI and the top atom are

$$J = \mathbb{E}_h[(1+Y)\ln(1+Y)], \qquad Q = J(a, 0), \qquad \boxed{A = J - Q - D + 2P.}$$

For $|Y| \le r < 1$, uniform absolute convergence permits integration of

$$(1+Y)\ln(1+Y) = Y + \sum_{n \ge 2} \frac{(-1)^n Y^n}{n(n-1)}.$$

A simultaneous shift of both angles by π negates Y, so odd moments vanish. The constant Fourier coefficient gives $\mathbb{E}\cos^{2j} v = \binom{2j}{j}/4^j$. Independence then yields

$$\begin{aligned}J &= \sum_{m \ge 1} \frac{M_m}{2m(2m-1)}, \\ M_m &= \frac{(2m)!}{4^m} \sum_{j=0}^{m} \frac{a^{2j} b^{2m-2j}}{(j!)^2((m-j)!)^2}.\end{aligned}$$

Every coefficient is nonnegative, including for negative amplitudes.

4 A positive denominator: all analytic steps

In $J(a,b) - J(a,0)$, the $j = m$ term cancels at each order. All remaining even-moment terms are nonnegative; the first is $b^2/4$. Therefore

$$J - Q \geq b^2/4.$$

A curvature check uses $C = \cos v$, $E = \cos u$ under the uniform reference. For t between zero and b, the positive floor $1 + aC + tE \geq 1 - r$ permits differentiation under the integral. Pairing (C, E) with $(-C, -E)$ gives

$$J_{bb}(a,t) = \mathbb{E}\frac{E^2}{1 + aC + tE} = \mathbb{E}\frac{E^2}{1 - (aC + tE)^2} \geq \mathbb{E}E^2 = \tfrac{1}{2}.$$

Independence and $\mathbb{E}E = 0$ give $J_b(a,0) = 0$, so

$$J(a,b) - J(a,0) = b^2 \int_0^1 (1-s) J_{bb}(a, sb)\, ds \geq b^2/4.$$

The integral remainder includes both signs of b.

For $0 \leq w \leq 1$, the floor $1 + awC \geq 1 - |a| > 0$ likewise justifies differentiating R. Algebra using $\mathbb{E}C = 0$ yields

$$R'(w) = a^2(1-w)\mathbb{E}\frac{C^2}{1 + awC},$$
$$R''(w) = -a^2\mathbb{E}\frac{(1 + aC)C^2}{(1 + awC)^2} \leq 0.$$

Thus R is concave. Since $R(0) = 0$ and $1/2 = (3/4)(2/3) + (1/4)0$,

$$P \geq \tfrac{3}{4}D, \qquad 2P - D \geq \tfrac{2}{3}P.$$

Symmetry of C gives

$$\mathbb{E}\frac{C^2}{1 + awC} = \mathbb{E}\frac{C^2}{1 - a^2w^2C^2} \geq \tfrac{1}{2}.$$

Consequently $R'(w) \geq a^2(1-w)/2$, and integration gives

$$P \geq \frac{a^2}{2}\left(\frac{1}{2} - \frac{1}{8}\right) = \frac{3a^2}{16}, \qquad 2P - D \geq a^2/8.$$

Combining the two contributions and completing a square,

$$\boxed{A \geq a^2/8 + b^2/4 \geq r^2/12,} \qquad \frac{a^2}{8} + \frac{b^2}{4} - \frac{r^2}{12} = \frac{(|a| - 2|b|)^2}{24} \geq 0.$$

No division by a, b, P, D occurred; their zero cases remain valid.

5 Uniform error and worked values

For integer $M \geq 1$, retain $m = 1, \ldots, M$ to define J_M. Since $M_m \leq r^{2m}$, replace every omitted denominator by $(2M+2)(2M+1)$ and sum a geometric series:

$$0 \leq J - J_M \leq B_M = \frac{r^{2M+2}}{(2M+2)(2M+1)(1-r^2)}.$$

Define $A_M = J_M - Q - D + 2P$, keeping the proper terms exact. Then $A - A_M = J - J_M \geq 0$. For $0 < r < 1$, the positive true denominator gives

$$0 \leq e_M := \frac{|A - A_M|}{A} \leq \frac{B_M}{a^2/8 + b^2/4} \leq \frac{12r^{2M}}{(2M+2)(2M+1)(1-r^2)}.$$

Both factors depending on r increase on $(0,1)$. Hence at $M = 8$, uniformly for $0 < r \leq 1/2$,

$$e_8 \leq \frac{12(1/2)^{16}}{18 \cdot 17 \cdot (3/4)} = \frac{1}{1253376} < \frac{1}{1000000}.$$

The final comparison is an integer inequality. It also implies $A_8 > 0$ off the origin. At the origin only zero absolute error is stated.

For $a = b = 1/4$, $r = 1/2$ and

$$A \geq \frac{3}{128}, \qquad B_8 = \frac{1}{60162048}, \qquad e_8 \leq \frac{1}{1410048}.$$

The first two truncations are $J_1 = 1/32$, $J_2 = 131/4096$, and

$$J_8 = \frac{3787665970322303}{1182194902184755 20},$$
$$A_8 = J_8 - H(1/4, 1/4) - H(1/4, 1/6) + 2H(1/4, 1/8).$$

Thus $[A_8, A_8 + 1/60162048]$ is a real-expression enclosure. It is not an executable outward-rounded interval.

a	b	Lower bound on A	Relative ceiling
1/2	0	1/32	1/1880064
0	1/2	1/16	1/3760128
1/4	1/4	3/128	1/1410048
1/3	1/6	1/48	1/1253376

All four cases have $r = 1/2$ and the same B_8. The last row attains only the weighted-square relaxation, not the actual atom or error bound. At $M = 7$, this same uniform ceiling is $1/245760 > 10^{-6}$; it does not certify the target and does not prove actual failure.

6 Negative controls and interpretation

A positive average does not imply a positive local atom. Set $a = 0$, $b = 1/4$, $u = \pi$. Every proper local cumulative is zero, but the local top atom is $\ln(3/4) < 0$. The average remains positive. Neither local nor averaged signs are silently clamped.

The top label is not added task information. At $b = 0$ and $a \neq 0$,

$$p(t \mid s_1, s_2, s_3) = p(t \mid s_1), \qquad I(S_2, S_3; T \mid S_1) = J - Q = 0,$$

while $A = 2P - D \geq a^2/8 > 0$. Ordinary MI/CMI answers the added-information question. A positive atom alone establishes no predictive, sensor-selection or control advantage.

MI's denominator cannot be silently reused. The derivative proof gives $Q > D > P > 0$ on the same $b = 0, a \neq 0$ axis. Therefore $A = 2P - D < P < Q = J$. A relative bound divided by J is not automatically a bound divided by A; the positive atom lower bound is the required additional argument.

The averaging law matters. Replacing $H(a, c)$ by $H(c, c)$ changes a proper-node value by $(a - c)c/(1 + \sqrt{1 - c^2})$, which is nonzero at $c = a/2, a \neq 0$. Multiplying all proper terms by b would wrongly remove their nonzero baseline at $b = 0$.

A tempting positive-coefficient proof fails. Expansion under the same bounded integral gives

$$R(w) = \sum_{m \geq 1} a^{2m} \frac{\binom{2m}{m}}{4^m} \left(\frac{w^{2m-1}}{2m - 1} - \frac{w^{2m}}{2m} \right).$$

The leading coefficient of $2P - D$ is $11/72$, but its a^6 coefficient is $-3089/2239488$. Its coefficients are not all positive. This rejects dropping its signed tail without proof; it does not by itself decide the stronger global inequality $2P - D \geq 11a^2/72$.

The boundary is excluded. At $r = 1$, the stated strict density floor and geometric denominator cease to apply. For $r > 1$, some phases give negative proposed density. Changed gauges, dependent or nonuniform sources, singular embeddings, rounded angles and other PID functionals require new arguments. The earlier mixed-dimensional raw-radius contrast remains unresolved; this analytic theorem does not repair an estimator.

7 Separate error budgets and legitimate use

The theorem approximates an exactly specified population law with known parameters. It does not estimate that law from data. The following error sources require separate evidence.

Source of error	Covered statement or open obligation
Series truncation	$0 \le A - A_M \le B_M$, with exact real Q, D, P.
Arithmetic	A computed $\widetilde{A}_M$ needs errors in J_M, Q, D, P; the theorem supplies none.
Parameter fitting	A recording-to-parameter map and its uncertainty are not supplied. Fitted values do not prove the population model.
Sampling/dependence	No IID premise is needed for the identities. Real phase time rows need their own temporal and observation assumptions.
Application	No held-out task gain, robustness, calibration or useful decision follows from the bound.

If qualified component enclosures later give absolute errors $\eta_J, \eta_Q, \eta_D, \eta_P$, the triangle inequality gives

$$|A - \widetilde{A}_M| \le B_M + \eta_J + \eta_Q + \eta_D + 2\eta_P.$$

For $r > 0$, division by $a^2/8 + b^2/4$ gives a conditional relative ceiling. These symbols do not assert that any such component enclosure has been implemented. Near the origin the true scale is quadratic in the amplitudes, so uncaptured absolute rounding error is especially relevant.

This is a useful offline reference for a deliberately specified stochastic phase channel: it tests the named decomposition, top-cut reconstruction, proper-node weighting and signed-atom interpretation. An analytic reference can reveal a formula mismatch even when finite samples look plausible. It cannot establish finite-sample calibration by itself.

Measured neural or interferometric phases would need justified phase extraction, gauge, grouping, target, noise model and joint law. No recording is shown to satisfy this two-harmonic density. No data result or performance measurement is reported. Two harmonic inputs are not two datasets.

For an actual decision, compare full MI/CMI, a known phase-residual statistic, direct task loss, fixed-model and retrained ablation, explicit failure utility, coverage, Bayesian or Fisher design, another PID and no PID. The chosen comparator must answer the same task. A short formula with a small truncation error does not justify a costly PID analysis by itself.

8 Rust implementation and computational cost

The exact input here is the supplied real pair (a, b), not a training set. Population support is the full four-dimensional angle box with periodic seams; there is no empirical alphabet or occupied-support count. A proposed sampled diagnostic would take N aligned rows of four scalar angles, three source columns and one target column. Parameter fitting, inference and resampling are additional operations outside this result.

No new Rust generator, series evaluator, outward interval, learner or binding is implemented by this report. The existing research route is `experimental::mixed_dimension_pid3::pid3_isx_report` and `pid3_isx`, in `crates/pid-core/src/pid3.rs`, under `research-mixed-dimension-pid3` with explicit runtime opt-in. Its sample-based full PID3 route retains mixed-dimensional raw-radius gaps. Incomplete PID3 is an availability diagnostic, not a missing-atom oracle.

The source-derived cost baseline records four triangular distance matrices: distance payload alone is $16N(N-1)$ bytes, before scratch and other allocations. Existing resource preflight and research warnings still apply. Its sample estimator is not this parameter-specified analytic calculation. No timing, peak-memory or real-time qualification is reported here.

The first M moments contain $M(M+3)/2$ nonnegative summands, forty-four at $M = 8$. Exact rational parameters make J_8 rational. A streaming evaluator may keep current coefficients, powers, moment and sum; caching powers and factorials uses $O(M)$ rational objects. Term counts are not constant-bit operation or memory bounds: numerators, denominators and GCD costs grow. Fixed-width factorial overflow, saturation or modular arithmetic cannot silently replace exact arithmetic.

The proper terms need qualified square-root and logarithm enclosures, plus subtraction-error propagation. A future evaluator needs explicit input bounds, resource limits, cancellation and fail-closed behavior. Those contracts are not supplied. Proof checking is separate from formula evaluation and estimator runtime; no formal kernel was run.

8.1 Evidence, source and citation

The detailed Markdown is *Relative truncation error for the averaged angular top atom* (5 October 2026). Its complete hand proof received an independent-first advisory model critique. Unchanged finite rational and 18-node checks passed in normal and optimized Python; they do not prove the integrals or continuum theorem. Exact retrieved-paper custody, formal replay, executable refinement and applications remain open.

Defining method: Ehrlich et al., *Partial Information Decomposition for Continuous Variables based on Shared Exclusions: Analytical Formulation and Estimation*, arXiv:2311.06373v3; Physical Review E 110, 014115 (2024). Software citation: pid-rs CITATION.cff. Suggested report citation: Sepehr Mahmoudian, *Relative error for an angular SxPID3 atom* (2026), including the exact repository commit or release when published. This private supplement is not accepted remote-main publication.